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## Potential as “intrinsic energy of inertia”

### Electro-Magnetic Field (force caused by potential)

- static: localized source charge at rest, only spatial dependency, no time dependency
- stationary: source charge at uniform movement, steady current flow, space and time dependency
- dynamic: source charge at fast movement, radiation, space and time dependency
- relativistic: source charge at very speedy movement, radiation, space and time dependency

- rest localization vs. time-dependency
- slow vs. fast movement
- vacuum vs. matter
- local vs. global of  $\mathcal{B}$ , singularity vs. regularity of  $\mathcal{E}$
- macro vs. micro
- far-field vs. near-field

### Magnetic Field in Vacuum

$$\mathcal{B} = \nabla \times U_m := \text{rot}(A) \stackrel{\text{Matter}}{=} \mu_0 (\mathcal{H} + \mathcal{M}) \stackrel{\text{Vacuum}}{=} \frac{1}{c} \left( \frac{r}{|r|} \times \mathcal{E} \right) \stackrel{\text{Far Field}}{=} \frac{1}{2} \frac{1}{2\pi} \mu_0 \frac{k^2 e^{i(kr - \omega t)}}{r} \frac{r}{|r|} \times \left( c p_{\text{dipol}} + m_0 \right) \quad (1)$$

### Magnetic Field in Matter

Excitation  $\hat{=}$  Displacement

$$\text{Excitation:} \quad \mathcal{H} = \frac{1}{\mu_0} \mathcal{B} - \mathcal{M} = \frac{1}{\mu} \mathcal{B} \quad \text{Magnetisation:} \quad \mathcal{M} = \frac{\mu - \mu_0}{\mu \mu_0} \mathcal{B} = \frac{\mu - \mu_0}{\mu_0} \mathcal{H} \quad (2)$$

## Electric Field in Vacuum

$$\mathcal{E} \stackrel{\text{conservativ}}{=} -\nabla U_e \stackrel{\text{dynamic}}{=} -\nabla U_e - \frac{\partial}{\partial t} U_m \stackrel{\text{fast moving charge}}{=} \frac{1}{2} \frac{1}{2\pi} \frac{1}{\varepsilon_0} \frac{1}{r} \frac{1}{\left(1 - \frac{r}{|r|} \cdot \frac{v}{c}\right)^3} q \left[ \underbrace{\left(1 - \frac{v^2}{c^2}\right) \frac{\frac{r}{|r|} - \frac{v}{c}}{r}}_{\text{uniform movement}} + \underbrace{\frac{\frac{r}{|r|} \times \left[\left(\frac{r}{|r|}\right) \cdot \frac{v}{c}\right]}{\text{accelerate}}}_{(3)} \right]$$

## Electric Field in Matter

Displacement  $\hat{=}$  Exitation (field change)

dielectricum = non-conducting material

electricum = conductor medium

$$\text{Exication:} \quad \mathcal{D} = \varepsilon_0 \mathcal{E} + \mathcal{P} = \stackrel{\text{isotropic medium in weak fields}}{=} \varepsilon_0 \mathcal{E} + \varepsilon_0 \chi \mathcal{E} = \varepsilon_0 (1 + \chi) \mathcal{E} = \varepsilon_0 4\pi r^3 n \langle p_m \rangle = 4\pi r^3 n \langle \mathcal{E}_m \rangle \quad (4)$$

## Abbreviations

$\Phi$  = electric potential       $A$  = magnetic potential       $j = \nabla \times \mathcal{H}$  = elec. current density

$\mathcal{E}$  = Electric Field       $\mathcal{B}$  = Magnetic Field       $\nabla \cdot \mathcal{B} = 0$        $\mathcal{D}$  = Displacement density of dielectricum       $\mathcal{M}$  = magnetization

$\mathcal{P}$  = Polarisation       $\mathcal{H}$  = magn. exitation, magn. field strength ("magn. displacement")

$n$  = spacial molecule density in volume

$\alpha = 4\pi r^3 = \text{constant}$        $\chi$  = elec. susceptibility of dielectricum       $\kappa = \frac{\mu - \mu_0}{\mu_0}$  = magn. susceptibility

$\langle p_m \rangle$  = spacial average of magn. dipol moment       $\langle \mathcal{E}_m \rangle$  = spacial average of electric field

$\bar{\mathcal{E}}_m$  = time average of electric field       $\mu$  = magn. permeability       $\mu_0$  = magn. constant ...

$\varepsilon$  = dielectricity constant       $\varepsilon_0$  = ...

## Electro-Magnetic interactions/coupling (Maxwell conditions)

**conservativ, static charge localisation:**

...

**stationary, slow charge movement:**

$$\nabla \cdot \varepsilon_0 \mathcal{E} = \varrho \quad \nabla \cdot \mathcal{B} = 0 \quad \nabla \times \mathcal{E} = -\frac{\partial}{\partial t} \mathcal{B} = i\omega \mathcal{B} \quad \nabla \times \frac{1}{\mu_0} \mathcal{B} = j + \varepsilon_0 \frac{\partial}{\partial t} \mathcal{E} = j - i\omega \varepsilon_0 \mathcal{E} \quad (5)$$

**relativistic, fast charge movement:**

...

## Potential derivation

⇒ Wave equation that determines Potential:

$$\Delta \cdot U - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} U = J \quad (6)$$

⇒ Potential caused by “ignition J” (inhomgenity) from source at rest,  $v = 0$ :

$$U = -\frac{1}{4\pi} \int \frac{J}{|r|} dr' \quad (7)$$

## elec. Potential structure

$$J = -\frac{1}{\varepsilon_0} \varrho$$

$$U_e := \Phi \stackrel{\text{elec. charge dens.}=\varrho}{=} \frac{1}{4\pi} \frac{1}{\varepsilon_0} \int \frac{\varrho}{|r|} dr' \hat{=} \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \frac{1}{\varepsilon_0} Q = \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \frac{1}{\varepsilon_0} It \quad (8)$$

## magn. Potential structure

$$J = -\mu_0 j$$

$$U_m := A \stackrel{\text{elec. current dens.}=j}{=} \frac{\mu_0}{4\pi} \int \frac{j}{|r|} dr' \hat{=} \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \mu_0 I = \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \frac{1}{c^2 \varepsilon_0} I = \frac{1}{c^2} \frac{1}{t} U_e \stackrel{\text{source in uniformal movement, } v=\text{const}}{=} \quad (9)$$

## Electric endogen energy

- Bond Energy
- Binding Energy

$$E_{pot} = mc^2$$

$$e = \text{electron charge} \quad q = ne = \text{charge} \quad n \in \mathbb{N} \quad (10)$$

$$\text{Potential} \quad U = \frac{E_{pot}}{q} = \frac{(mc^2)^2}{ne} = \frac{m}{q} mc^4 = \kappa mc^4 \quad (11)$$

## Electric induced energy (exogenous excitation)

- Ionisation Energy
- Excitation Energy

$$E_{kin} = \frac{1}{2} mv^2 \hat{=} (pc)^2 \hat{=} h\nu = \hbar\omega$$

$E_{kin}$  Energy density electric:

$$w_e = \frac{1}{2} \varepsilon_0 \mathcal{E}^2 \quad (12)$$

$E_{kin}$  Energy density magnetic:

$$w_m = \frac{1}{2} \frac{1}{\mu_0} \mathcal{B}^2 \quad (13)$$

$E_{kin}$  Electric Energy:

$$W_e = \frac{1}{2} CU^2 \quad (14)$$

$E_{kin}$  Magnetic Energy:

$$W_m = \frac{1}{2} LI^2 \quad (15)$$

**Power:**

$$P = VI = (U_1 - U_2) \frac{Q}{t} = \frac{E_1 - E_2}{t} = \frac{W}{t} = \frac{Fr}{t} = Fv \quad (16)$$

**Frequency:**

$$\omega = \dot{\rho} = \frac{v}{r} = 2\pi\nu = k\lambda \frac{1}{T} = \sqrt{\frac{k}{m}} \stackrel{vacuum}{=} kc \stackrel{matter}{=} k \frac{c}{\eta} = kc \frac{\tan \beta}{\tan \alpha} \stackrel{electromagnetic}{=} kc \left( \mathcal{E}_\beta^{-1} \mathcal{E}_\alpha \right)^{-1} = kc \frac{\varepsilon_\beta}{\varepsilon_\alpha} = kc \sqrt{\frac{\varepsilon_0 \mu}{\varepsilon \mu}} \quad (17)$$

**Phase:**

$$\varphi = k \left( \vec{r} \cdot \vec{k} \right) - \omega t := ks - \omega t = \dots \quad (18)$$

**Phaseshift:**

$$\Delta\varphi = \Delta k \left( \vec{r} \cdot \vec{\Delta k} \right) - \Delta\omega t := \Delta ks - \Delta\omega t = \dots \quad (19)$$

**Refraction:**

Refraction  $\Leftrightarrow \exists$  Matter; refraction causes dispersion

$$\eta = \frac{\tan \alpha}{\tan \beta} = \frac{\lambda_0}{\lambda} = c\sqrt{\varepsilon\mu} \hat{=} \mathcal{E}_\beta^{-1} \mathcal{E}_\alpha = \frac{\varepsilon_\alpha}{\varepsilon_\beta} = \dots \quad (20)$$

**Dispersion:**

Dispersion  $\Leftrightarrow \exists$  Matter (dielectric medium or conductor); Dispersion  $\neq$  Dissipation  $\neq$  Diffusion

$$\dots \quad (21)$$

**Dielectricity:**

$$\varepsilon(\omega) = \dots \quad (22)$$

**Permeability:**

$$\mu(\omega) = \dots \quad (23)$$

### Conductivity:

$$\sigma(\omega) = \dots \quad (24)$$

### Conductor (charge conductive element for energy transfer) condition:

$$\sigma \cdot t \gg \varepsilon_0 \quad (25)$$

### Slow changing current (quasi-stationary, low frequencies) condition:

$$\frac{v}{c} \ll \frac{1}{c} \frac{\mathcal{E}}{\mathcal{B}} \ll \frac{c}{v} \quad (26)$$

## Energy Balance

Energy Balance

$$\frac{\partial}{\partial t} \sum w_i + \cdot \sum \mathcal{S}_{w_i} + j \cdot \mathcal{E} = 0 \quad (27)$$

Abbreviations:

$w_i$  = engery densities (electric, magnetic, gravitational, etc. ...),  $i \in \mathbb{N}$

$\mathcal{S}_{w_i}$  = Poynting-Vectros  $\hat{=}$  energy transport densities (transfers) by fields (electric, magnetic, gravitational, etc. ...) to corresponding energy densities  $w_i$

$j \cdot \mathcal{E}$  = energy gain provided by fields  $\mathcal{E}$  and  $\mathcal{B}$  through current density  $j$

## Mechanical

...

## Electromagnetic energy transfer

charge conservation:

$$\frac{\partial}{\partial t} \varrho + \nabla \cdot j = 0 \quad (28)$$

$\varrho = \sigma \delta$  = charge density in interior

$\sigma$  = charge density at border (surface)

$\delta$  = Dirac delta-function

$j = \sigma (\mathcal{E} + v \times \mathcal{B}) = \nabla \times \mathcal{H} - \frac{\partial}{\partial t} \mathcal{D} \hat{=}$  electric current density

$j \cdot \mathcal{E} \hat{=}$  energy gain by fields  $\mathcal{E}$  and  $\mathcal{B}$  through current density  $j$

$\mathcal{S} = \mathcal{E} \times \mathcal{H} \hat{=}$  energy transport density (Intensity =  $|\bar{\mathcal{S}}|$ )

Energy conservation for (large extended) conductive medium:

$$\int \left( \frac{\partial}{\partial t} (w_e + w_m) + \cdot \mathcal{S} + j \cdot \mathcal{E} \right) dr = \frac{d}{dt} W_e + \frac{d}{dt} W_m + D_S + \int (j \cdot \mathcal{E}) dr \stackrel{\text{low frequency}}{=} \frac{1}{2} \frac{d}{dt} (LI^2) + \frac{1}{2} \frac{d}{dt} \left( \frac{Q^2}{C} \right) \quad (29)$$

Abbreviations:

$$\Delta \cdot := \text{div grad} = \text{Laplacian}$$

$J$  = "ignition of inhomogeneity, cause"     $j$  = Current density     $\varrho$  = Charge density

$V$  = Potential difference     $I$  = Current     $R$  = Resistance

$L$  = Inductance     $C$  = Capacitance     $Q$  = charge (ion)     $\Psi = LI$  = mag. Flux

$\mathcal{S} = \mathcal{E} \times \mathcal{H}$  = Poynting-Vector, radiation intensity

$D_S = \int (\cdot \mathcal{S}) dr = \int \text{div}(\mathcal{E} \times \mathcal{H}) dr$  = dissipation energy lost from radiation  $\approx 0$  for low frequencies  $\omega$



$$d = \sqrt{\frac{1}{\mu\sigma\omega}} = \text{skin depth (surface) of conductor} \quad r = \frac{d}{\sqrt{2i}}\rho = \frac{d}{1+i}\rho = \text{radius (extension) of conductor}$$

$$\rho = kr = \text{"radius of curvature } k\text{"}$$

$$\mu = \text{permeability of conductor} \quad \sigma = \text{conductivity of conductor (dielectricum)}$$

### Thermal

...

### Gravitational

...

**Energy of thin conductor (concentrated conductive medium without electric charge in its surrounding), Voltage (potential difference, tension):**

$$V = U_1 - U_2 = r\mathcal{E} = \dot{\Psi} + RI + \frac{I}{C}t = L\dot{I} + RI + \frac{Q}{C} \quad (30)$$

### Low Frequency

$$\omega \ll \frac{1}{\mu\sigma} \frac{2}{r^2} \iff r \ll d \iff \rho \ll 1 \quad (31)$$

$$L \approx \text{const.} \quad C \approx \text{const.} \quad D_S \approx 0$$

## High Frequency

$$\omega \gg \frac{1}{\mu\sigma} \frac{2}{r^2} \iff r \gg d \iff \rho \gg 1 \quad (32)$$
$$L(t) \neq \text{const.} \quad C(t) \neq \text{const.} \quad \text{for Dipol: } D_S \propto \omega^4 \gg 0$$

## Continuity at Rest: Electric Charge

no Dissipation, no velocity, charge (ion) causes Electric Field

## Continuity at Movement: Magnetic Field

no Dissipation, Velocity constant, moved charge (ion) causes emergence of Magnetic Field

## Continuity at Acceleration: Radiation Energy

no Dissipation, Velocity changing, acceleration of charge (ion) causes emergence of Radiation

## Continuity and periodic (harmonious oscillation) Movement in VACUUM

$$c = \frac{1}{\sqrt{\varepsilon_0 \mu_0}} = \nu \lambda = \frac{\lambda}{T}$$
$$2\pi = k\lambda = T\omega = \frac{\omega}{\nu}$$

$$2\pi = k\lambda \gg kr_0 \ll 1$$

$$\omega = kc \ll \frac{c}{r_0} \quad \nabla \cdot j = i\omega \varrho \quad i\omega Q = 0$$

## Continuity and periodic (harmonious oscillation) Movement in MATTER

Refraction index  $\eta$  emerges:  $\varepsilon\mu = \left(\frac{\eta}{v_{ph}}\right)^2 \hat{=} \left(\frac{\eta}{c}\right)^2$

Dispersion:  $k^2 = \varepsilon\mu\omega^2 + i\mu\sigma\omega$

## Near Field

static, vacuum, granular spacetime

$$r \ll \lambda = \frac{2\pi}{k} \stackrel{\text{vacuum}}{=} \frac{2\pi}{\frac{\omega}{c}} \quad kr \ll 2\pi$$

$$\frac{w_m}{w_e} \sim \frac{\frac{\vec{p}^2}{\epsilon_0 r^6}}{\frac{k^2 \mu_0 c^2 \vec{p}^2}{r^4}} \sim (kr)^2 \ll 1$$

$\vec{p}$  = elec. Dipol

## Far Field

static, vacuum, flat spacetime

accelerated movement -> Radiation in Far Field of charge source

$$r \gg \lambda = \frac{2\pi}{k} \stackrel{\text{vacuum}}{=} \frac{2\pi}{\frac{\omega}{c}} \quad kr \gg 1 \quad r \gg kr_0^2 \ll 1$$

$$\frac{w_m}{w_e} = \frac{\frac{\epsilon_0}{2} c^2 \left( \mathcal{B} \times \frac{r}{|r|} \right)^2}{\frac{\epsilon_0}{2} c^2 \left( \mathcal{B} \times \frac{r}{|r|} \right)^2} = 1$$

## Dissipation in Vacuum

no Continuity of charge/matter, no Refraction index, Reflection

## Dissipation in Matter

no Continuity of charge/matter with dielectricum, refraction merges, reflection, absorption

## Energy gain by excitation

Action Quantum, Planck Constant (Energy-Frequency-Slope)<sup>1</sup>:

---

<sup>1</sup>Wirkungsquantum

$$h = \tan(\varphi) = \frac{E_{kin}}{\nu_i - \nu_0} = \frac{eV_{0i}}{\nu_i - \nu_0} \equiv \frac{E_1 - E_2}{\nu_0} = \frac{W}{\nu_0} = \frac{F_{EM} \cdot x}{\nu_0} = \hbar k \lambda \quad (33)$$

$$\hbar = \frac{E_1 - E_2}{\omega} = \quad (34)$$

$$E_{kin}^{max} = eV_0 = e \frac{Q}{C} = h(\nu - \nu_0) \quad (35)$$

$V_0$  = Opposing Potential ( $I = 0$ )

$\varphi = \angle(E_{kin}, \nu)$        $e$  = electron charge       $Q$  = ion charge

$$C = \text{Capacitance} \quad \nu = \frac{1}{2\pi} \frac{1}{\sqrt{LC}} \sqrt{1 - \frac{C}{L} R^2}$$

$$\Rightarrow \text{Inductance } L = \frac{4\pi^2}{C} \left( \frac{e}{h} Q + \nu_0 \right)^2$$

$\Rightarrow$  Ionisation Energy of chemical element  $E_i = h\nu = h \frac{\lambda}{c} = \hbar \omega = \frac{h}{k\lambda} \omega$

$\Rightarrow$  Energy released (gained):

- with mass (e.g. electron), Tunnel-Effect:  $E_{gain} = E_{pot} - E_i = mc^2 - h\nu$
- massless (e.g. photon), Doppler-Effect (frequency shift):  $E_{gain} = E_{kin} - E_i = h(\nu_0 - \nu_i) = \Delta\omega\hbar$

## Frequency Cause-Effect

### Frequency (commonness, probability, likelihood)

$$\omega = \frac{d}{dt}\varphi = \eta kv = kc = \frac{2\pi}{\mathcal{T}} = 2\pi \cdot \nu = \frac{a}{v} = \frac{v}{r} = \frac{E}{\hbar} = \sqrt{\frac{k}{m}} = \sqrt{\frac{r \times F}{m}} = \sqrt{\frac{g}{l}} \stackrel{\text{electromagnetic}}{=} \dots = \sqrt{\frac{1}{LC}} \stackrel{QM}{=} \frac{1}{2} \frac{p^2}{m \hbar} \quad (36)$$

$$\nu = \frac{E}{h} = \frac{v_{phs}}{\lambda} = \frac{c^2}{v} \frac{1}{\lambda} = \frac{\omega}{k} \frac{1}{\lambda} = \frac{\omega}{2\pi} \quad (37)$$

$$\nu_i = \frac{eV_{0i}}{h} + \nu_0 = \frac{E_{kin}}{\hbar} \omega \mathcal{T} + \nu_0 \quad (38)$$

$$e = \text{electron charge} \quad V_0 = \text{Opposing Potential } (I = 0) \quad (39)$$

Energy Level:  $i = [1, n] \in \mathbb{N}$

$$E_{kin}^{max} = eV_0 = h(\nu - \nu_0) \quad (40)$$

### Frequency contextual Causes

Frequency of rotational velocity as periodic occurrence of an event/state by space distance and time periodicity as well as movement speed and reach length,  $v^2 = v_{ph} v_{gr}$ ,  $\omega t = 2\pi \nu t = \lambda \neq 0$  :

$$\frac{v}{r} = \omega = \frac{\lambda}{t} \quad (41)$$

### Rest state of source charge

Statics localization of charge. Only spacial demependency, no time dependency.

Energy transport ... Information transmission ...

$$v = 0 \quad \dot{v} = 0$$

### Slow Movement of source charge

Stationarity localization of current flow by slow movement of charge. Dynamics, time dependency.

Energy transport ... Information transmission ...

$$v = \text{const.} \ll c \quad \dot{v} = 0$$

### Fast Movement of source charge

Fields forces by fast movement of charge. Dynamics, time dependency.

Energy transport ... Information transmission ...

$$v \approx c \quad \dot{v} = 0$$

### Acceleration Movement of source charge

Radiation energy by accelerated movement of charge. Dynamics, time dependency.

Energy transport ... Information transmission ...

$$v \neq 0 \quad \dot{v} \neq 0$$

### Movement in Vacuum

$$\eta = \frac{\tan \alpha}{\tan \beta} = \frac{c}{v_{ph}} = 1 \quad 2\pi = k\lambda \quad v_{ph} = c \quad \omega = kc$$

$$\varepsilon_0 \mu_0 = \frac{1}{c^2}$$

$$\text{Continuity: } \nabla \cdot j = i\omega \varrho \quad i\omega Q = 0$$

### Movement in Matter

$$\eta = \frac{\tan \alpha}{\tan \beta} = \frac{c}{v_{ph}} = \sqrt{\frac{\varepsilon\mu}{\varepsilon_0\mu_0}} \neq 1 \quad v_{ph} = \pm \sqrt{\frac{1}{\varepsilon\mu} - \frac{\sigma^2}{4\varepsilon^2 k^2}} \quad \omega = kv_{ph} = kc \sqrt{\frac{\varepsilon_0\mu_0}{\varepsilon\mu}} =$$

$$kc \frac{\tan \beta}{\tan \alpha} = k \frac{c}{\eta}$$

$$\varepsilon\mu = \left(\frac{\eta}{c}\right)^2$$

$$\text{Dispersion: } k^2 = \varepsilon\mu\omega^2 + i\mu\sigma\omega$$

### Near Field

$$r_0 \ll r \ll \lambda \quad kr \ll 2\pi \quad \frac{w_m}{w_e} \ll 1$$

$r$  is small, fast movement

## Far Field

$$r \gg \lambda \quad kr \gg 1 \quad r \gg kr_0^2 \ll 1 \quad \frac{w_m}{w_e} = 1$$

$r$  is big, slow movement

## High Frequency

$$\omega \gg \frac{2}{\mu\sigma r_0^2}$$

$\lambda$  is big,  $\nu$  frequency is small, short time period

## Low Frequency

$$\omega \ll \frac{2}{\mu\sigma r_0^2}$$

$\lambda$  is small,  $\nu$  frequency is big, long time period

## Conditions

Border (skin surface): ...

Interior (medium body): ...

Global: ....

Local: ....

## Consequences

dielectricity constant (matter medium):  $\varepsilon = \varepsilon(\omega) = \mathcal{E}^{-1} \mathcal{D} = \mathcal{E}^{-1} (\mathcal{P} + \varepsilon_0 \mathcal{B})$

permeability constant (matter medium):  $\mu = \mu(\omega) = \mathcal{B}^{-1} \mathcal{H} = \mathcal{B}^{-1} \left( \frac{1}{\mu_0} \mathcal{B} - \mathcal{M} \right)$

conductivity (matter medium):  $\sigma = \sigma(\omega) = \frac{\sigma_0}{\sqrt{1 + \omega^2 \tau^2}}$

magn. energy density (vacuum):  $w_m = \frac{1}{2} \frac{1}{\mu_0} \mathcal{B}^2$

elec. energy density (vacuum):  $w_e = \frac{1}{2} \varepsilon_0 \mathcal{E}^2$

wave vector:  $k = \frac{2\pi}{\lambda} = \eta \frac{\omega}{c}$

refraction index (matter medium):  $\eta = \frac{\tan \alpha}{\tan \beta} = \frac{\eta_{medium1}}{\eta_{medium2}}$

Energy:  $E = h\nu = \hbar\omega = E_{kin} + E_{pot} \stackrel{!}{=} \sqrt{(pc)^2 + (mc^2)^2}$

## Operators

Operators

$\nabla := grad$  = Gradient (Gefälle, Neigung)

$\nabla \cdot := div$  = Divergence (Quelle)

$\nabla \times := rot$  = Rotation (Wirbelung, Curl)

$\nabla^2$  = Lagrange Operator (Krümmung, Beschleunigung)

$\Delta \cdot := div(grad)$  = Laplace Operator

$\Delta$  := Difference, shift